Introduction to Java

What is Java

Java is a **programming language** and a **platform**. Java is a high level, robust, object-oriented and secure programming language.

Java was developed by *Sun Microsystems* (which is now the subsidiary of Oracle) in the year 1995. *James Gosling* is known as the father of Java. Before Java, its name was *Oak*. Since Oak was already a registered company, so James Gosling and his team changed the Oak name to Java.

Platform: Any hardware or software environment in which a program runs, is known as a platform. Since Java has a runtime environment (JRE) and API, it is called a platform.

Java Example

Let's have a quick look at Java programming example. A detailed description of Hello Java example is available in next page.

```
class Simple{
    public static void main(String args[]){
     System.out.println("Hello Java");
    }
}
```

Application

According to Sun, 3 billion devices run Java. There are many devices where Java is currently used. Some of them are as follows:

1. Desktop Applications such as acrobat reader, media player, antivirus, etc.
2. Web Applications such as irctc.co.in, javatpoint.com, etc.
3. Enterprise Applications such as banking applications.
4. Mobile
5. Embedded System
6. Smart Card
7. Robotics
8. Games, etc.

Types of Java Applications

There are mainly 4 types of applications that can be created using Java programming:

1) Standalone Application

Standalone applications are also known as desktop applications or window-based applications. These are traditional software that we need to install on every machine. Examples of standalone application are Media player, antivirus, etc. AWT and Swing are used in Java for creating standalone applications.

2) Web Application

An application that runs on the server side and creates a dynamic page is called a web application. Currently, Servlet, JSP, Struts, Spring, Hibernate, JSF, etc. technologies are used for creating web applications in Java.

3) Enterprise Application

An application that is distributed in nature, such as banking applications, etc. is called enterprise application. It has advantages of the high-level security, load balancing, and clustering. In Java, EJB is used for creating enterprise applications.

4) Mobile Application

An application which is created for mobile devices is called a mobile application. Currently, Android and Java ME are used for creating mobile applications.

Java Platforms / Editions

There are 4 platforms or editions of Java:

1) Java SE (Java Standard Edition)

It is a Java programming platform. It includes Java programming APIs such as java.lang, java.io, java.net, java.util, java.sql, java.math etc. It includes core topics like OOPs, String, Regex, Exception, Inner classes, Multithreading, I/O Stream, Networking, AWT, Swing, Reflection, Collection, etc.

2) Java EE (Java Enterprise Edition)

It is an enterprise platform which is mainly used to develop web and enterprise applications. It is built on the top of the Java SE platform. It includes topics like Servlet, JSP, Web Services, EJB, JPA, etc.

3) Java ME (Java Micro Edition)

It is a micro platform which is mainly used to develop mobile applications.

4) JavaFX

It is used to develop rich internet applications. It uses a light-weight user interface API.

History of Java

The history of Java is very interesting. Java was originally designed for interactive television, but it was too advanced technology for the digital cable television industry at the time. The history of Java starts with the Green Team. Java team members (also known as **Green Team**), initiated this project to develop a language for digital devices such as set-top boxes, televisions, etc. However, it was suited for internet programming. Later, Java technology was incorporated by Netscape.

The principles for creating Java programming were "Simple, Robust, Portable, Platform-independent, Secured, High Performance, Multithreaded, Architecture Neutral, Object-Oriented, Interpreted, and Dynamic". Java was developed by James Gosling, who is known as the father of Java, in 1995. James Gosling and his team members started the project in the early '90s.

Currently, Java is used in internet programming, mobile devices, games, e-business solutions, etc. There are given significant points that describe the history of Java.

1) **James Gosling**, **Mike Sheridan**, and **Patrick Naughton** initiated the Java language project in June 1991.
The small team of sun engineers called **Green Team**.
2) Initially designed for small, embedded systems in electronic appliances like set-top boxes.
3) Firstly, it was called **"Greentalk"** by James Gosling, and the file extension was .gt.
4) After that, it was called **Oak** and was developed as a part of the Green project.

Why Java named "Oak"?

Why Oak? Oak is a symbol of strength and chosen as a national tree of many countries like the U.S.A., France, Germany, Romania, etc.

In 1995, Oak was renamed as **"Java"** because it was already a trademark by Oak Technologies.

Why Java Programming named "Java"?

Why had they chosen java name for java language? The team gathered to choose a new name. The suggested words were "dynamic", "revolutionary", "Silk", "jolt", "DNA", etc. They wanted something that reflected the essence of the technology: revolutionary, dynamic, lively, cool, unique, and easy to spell and fun to say.

According to James Gosling, "Java was one of the top choices along with **Silk**". Since Java was so unique, most of the team members preferred Java than other names.

Java is an island of Indonesia where the first coffee was produced (called java coffee). It is a kind of espresso bean. Java name was chosen by James Gosling while having coffee near his office.

Notice that Java is just a name, not an acronym.

Initially developed by James Gosling at Sun Microsystems (which is now a subsidiary of Oracle Corporation) and released in 1995.

In 1995, Time magazine called **Java one of the Ten Best Products of 1995**.
JDK 1.0 released in(January 23, 1996). After the first release of Java, there have been many additional features added to the language. Now Java is being used in Windows applications, Web applications, enterprise applications, mobile applications, cards, etc. Each new version adds the new features in Java.

Java Version History

Many java versions have been released till now. The current stable release of Java is Java SE 10.

1. JDK Alpha and Beta (1995)
2. JDK 1.0 (23rd Jan 1996)
3. JDK 1.1 (19th Feb 1997)
4. J2SE 1.2 (8th Dec 1998)
5. J2SE 1.3 (8th May 2000)
6. J2SE 1.4 (6th Feb 2002)
7. J2SE 5.0 (30th Sep 2004)
8. Java SE 6 (11th Dec 2006)
9. Java SE 7 (28th July 2011)
10. Java SE 8 (18th Mar 2014)
11. Java SE 9 (21st Sep 2017)
12. Java SE 10 (20th Mar 2018)

Features of Java

The primary objective of Java programming language creation was to make it portable, simple and secure programming language. Apart from this, there are also some excellent features which play an important role in the popularity of this language. The features of Java are also known as java *buzzwords*.
A list of most important features of Java language is given below.

1. Simple
2. Object-Oriented
3. Portable
4. Platform independent
5. Secured
6. Robust
7. Architecture neutral
8. Interpreted
9. High Performance
10. Multithreaded
11. Distributed
12. Dynamic

Simple

Java is very easy to learn, and its syntax is simple, clean and easy to understand. According to Sun, Java language is a simple programming language because:

- Java syntax is based on C++ (so easier for programmers to learn it after C++).
- Java has removed many complicated and rarely-used features, for example, explicit pointers, operator overloading, etc.
- There is no need to remove unreferenced objects because there is an Automatic Garbage Collection in Java.

Object-oriented

Java is an object-oriented programming language. Everything in Java is an object. Object-oriented means we organize our software as a combination of different types of objects that incorporates both data and behavior. Object-oriented programming (OOPs) is a methodology that simplifies software development and maintenance by providing some rules.

Basic concepts of OOPs are:

1. Object
2. Class
3. Inheritance
4. Polymorphism
5. Abstraction
6. Encapsulation

Platform Independent

Java is platform independent because it is different from other languages like C, C++, etc. which are compiled into platform specific machines while Java is a write once, run anywhere language. A platform is the hardware or software environment in which a program runs.

There are two types of platforms software-based and hardware-based. Java provides a software-based platform.

The Java platform differs from most other platforms in the sense that it is a software-based platform that runs on the top of other hardware-based platforms. It has two components:

1. Runtime Environment
2. API(Application Programming Interface)

Java code can be run on multiple platforms, for example, Windows, Linux, Sun Solaris, Mac/OS, etc. Java code is compiled by the compiler and converted into bytecode. This bytecode is a platform-independent code because it can be run on multiple platforms, i.e., Write Once and Run Anywhere(WORA).

Secured

Java is best known for its security. With Java, we can develop virus-free systems. Java is secured because:

- **No explicit pointer**
- **Java Programs run inside a virtual machine sandbox**
- **Classloader:** Classloader in Java is a part of the Java Runtime Environment(JRE) which is used to load Java classes into the Java Virtual Machine dynamically. It adds security by separating the package for the classes of the local file system from those that are imported from network sources.
- **Bytecode Verifier:** It checks the code fragments for illegal code that can violate access right to objects.
- **Security Manager:** It determines what resources a class can access such as reading and writing to the local disk.

Java language provides these securities by default. Some security can also be provided by an application developer explicitly through SSL, JAAS, Cryptography, etc.

Robust

Robust simply means strong. Java is robust because:

- It uses strong memory management.
- There is a lack of pointers that avoids security problems.
- There is automatic garbage collection in java which runs on the Java Virtual Machine to get rid of objects which are not being used by a Java application anymore.
- There are exception handling and the type checking mechanism in Java. All these points make Java robust.

Architecture-neutral

Java is architecture neutral because there are no implementation dependent features, for example, the size of primitive types is fixed.

In C programming, int data type occupies 2 bytes of memory for 32-bit architecture and 4 bytes of memory for 64-bit architecture. However, it occupies 4 bytes of memory for both 32 and 64-bit architectures in Java.

Portable

Java is portable because it facilitates you to carry the Java bytecode to any platform. It doesn't require any implementation.

High-performance

Java is faster than other traditional interpreted programming languages because Java bytecode is "close" to native code. It is still a little bit slower than a compiled language (e.g., C++). Java is an interpreted language that is why it is slower than compiled languages, e.g., C, C++, etc.

Distributed

Java is distributed because it facilitates users to create distributed applications in Java. RMI and EJB are used for creating distributed applications. This feature of Java makes us able to access files by calling the methods from any machine on the internet.

Multi-threaded

A thread is like a separate program, executing concurrently. We can write Java programs that deal with many tasks at once by defining multiple threads. The main advantage of multi-threading is that it doesn't occupy memory for each thread. It shares a common memory area. Threads are important for multi-media, Web applications, etc.

Dynamic

Java is a dynamic language. It supports dynamic loading of classes. It means classes are loaded on demand. It also supports functions from its native languages, i.e., C and C++.

Java supports dynamic compilation and automatic memory management (garbage collection).

C++ vs Java

There are many differences and similarities between the C++ programming language and Java. A list of top differences between C++ and Java are given below:

Comparison Index	C++	Java
Platform-independent	C++ is platform-dependent.	Java is platform-independent.
Mainly used for	C++ is mainly used for system programming.	Java is mainly used for application programming. It is widely used in window, web-based, enterprise and mobile applications.
Design Goal	C++ was designed for systems and applications programming. It was an extension of C programming language.	Java was designed and created as an interpreter for printing systems but later extended as a support network computing. It was designed with a goal of being easy to use and accessible to a broader audience.
Goto	C++ supports the goto statement.	Java doesn't support the goto statement.
Multiple inheritance	C++ supports multiple inheritance.	Java doesn't support multiple inheritance through class. It can be achieved by interfaces in java.
Operator	C++ supports operator overloading.	Java doesn't support operator overloading.

Overloading		
Pointers	C++ supports pointers. You can write pointer program in C++.	Java supports pointer internally. However, you can't write the pointer program in java. It means java has restricted pointer support in java.
Compiler and Interpreter	C++ uses compiler only. C++ is compiled and run using the compiler which converts source code into machine code so, C++ is platform dependent.	Java uses compiler and interpreter both. Java source code is converted into bytecode at compilation time. The interpreter executes this bytecode at runtime and produces output. Java is interpreted that is why it is platform independent.
Call by Value and Call by reference	C++ supports both call by value and call by reference.	Java supports call by value only. There is no call by reference in java.
Structure and Union	C++ supports structures and unions.	Java doesn't support structures and unions.
Thread Support	C++ doesn't have built-in support for threads. It relies on third-party libraries for thread support.	Java has built-in thread support.
Documentation comment	C++ doesn't support documentation comment.	Java supports documentation comment (/** ... */) to create documentation for java source code.
Virtual Keyword	C++ supports virtual keyword so that we can decide whether or not override a function.	Java has no virtual keyword. We can override all non-static methods by default. In other words, non-static methods are virtual by default.
unsigned right shift >>>	C++ doesn't support >>> operator.	Java supports unsigned right shift >>> operator that fills zero at the top for the negative numbers. For positive numbers, it works same like >> operator.
Inheritance Tree	C++ creates a new inheritance tree always.	Java uses a single inheritance tree always because all classes are the child of Object class in java. The object class is the root of the inheritance tree in java.
Hardware	C++ is nearer to hardware.	Java is not so interactive with hardware.
Object-oriented	C++ is an object-oriented language. However, in C language, single root hierarchy is not possible.	Java is also an object-oriented language. However, everything (except fundamental types) is an object in Java. It is a single root hierarchy as everything gets derived from java.lang.Object.

Note

- Java doesn't support default arguments like C++.
- Java does not support header files like C++. Java uses the import keyword to include different classes and methods.

C++ Example

File: main.cpp

```
#include <iostream>
using namespace std;
int main() {
   cout << "Hello C++ Programming";
   return 0;
}
```

Java Example

File: Simple.java

```
class Simple{
    public static void main(String args[]){
     System.out.println("Hello Java");
    }
}
```

First Java Program | Hello World Example

Here, we will learn how to write the simple program of java. We can write a simple hello java program easily after installing the JDK.

To create a simple java program, you need to create a class that contains the main method. Let's understand the requirement first.

The requirement for Java Hello World Example

For executing any java program, you need to

- Install the JDK if you don't have installed it, download the JDK and install it.
- Set path of the jdk/bin directory.
- Create the java program
- Compile and run the java program

Creating Hello World Example

Let's create the hello java program:

```
class Simple{
    public static void main(String args[]){
     System.out.println("Hello Java");
    }
}
```

save this file as Simple.java

To compile: javac Simple.java

To execute: java Simple

Output: Hello Java

Compilation Flow:

When we compile Java program using javac tool, java compiler converts the source code into byte code.

Parameters used in First Java Program

Let's see what is the meaning of class, public, static, void, main, String[], System.out.println().

- **class** keyword is used to declare a class in java.
- **public** keyword is an access modifier which represents visibility. It means it is visible to all.
- **static** is a keyword. If we declare any method as static, it is known as the static method. The core advantage of the static method is that there is no need to create an object to invoke the static method. The main method is executed by the JVM, so it doesn't require to create an object to invoke the main method. So it saves memory.
- **void** is the return type of the method. It means it doesn't return any value.
- **main** represents the starting point of the program.
- **String[] args** is used for command line argument. We will learn it later.
- **System.out.println()** is used to print statement. Here, System is a class, out is the object of PrintStream class, println() is the method of PrintStream class.

How many ways can we write a Java program

There are many ways to write a Java program. The modifications that can be done in a Java program are given below:

1) By changing the sequence of the modifiers, method prototype is not changed in Java.

Let's see the simple code of the main method.

```
static public void main(String args[])
```

2) The subscript notation in Java array can be used after type, before the variable or after the variable.

Let's see the different codes to write the main method.

```
public static void main(String[] args)
public static void main(String []args)
public static void main(String args[])
```

3) You can provide var-args support to the main method by passing 3 ellipses (dots)

Let's see the simple code of using var-args in the main method. We will learn about var-args later in Java New Features chapter.

```
public static void main(String... args)
```

4) Having a semicolon at the end of class is optional in Java.

Let's see the simple code.

```
class A{
   static public void main(String... args){
        System.out.println("hello java4");
   }
};
```

Valid java main method signature

```
public static void main(String[] args)
public static void main(String []args)
public static void main(String args[])
public static void main(String... args)
static public void main(String[] args)
public static final void main(String[] args)
final public static void main(String[] args)
final strictfp public static void main(String[] args)
```

Invalid java main method signature

```
public void main(String[] args)
static void main(String[] args)
public void static main(String[] args)
abstract public static void main(String[] args)
```

Resolving an error "javac is not recognized as an internal or external command"?

If there occurs a problem like displayed in the below figure, you need to set path. Since DOS doesn't know javac or java, we need to set path. The path is not required in such a case if you save your program inside the JDK/bin directory. However, it is an excellent approach to set the path.

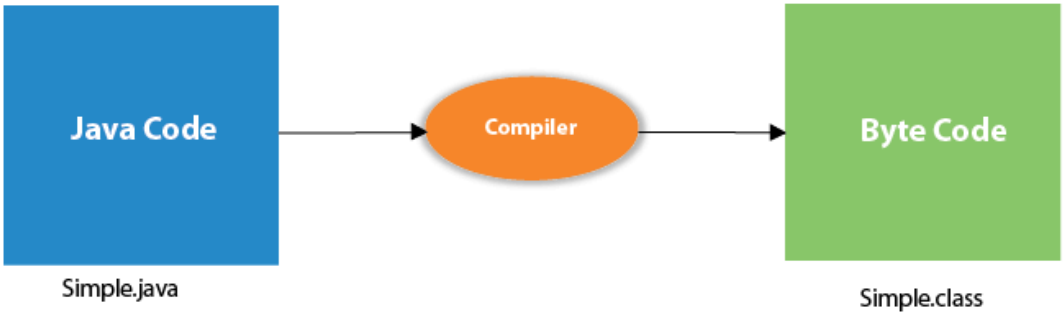
Internal Details of Hello Java Program

Internal Details of Hello Java

In the previous page, we have learnt about the first program, how to compile and run the first java program. Here, we are going to learn, what happens while compiling and running the java program. Moreover, we will see some question based on the first program.

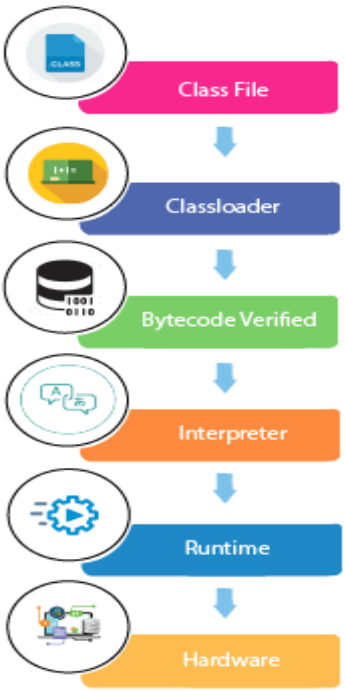
What happens at compile time?

At compile time, java file is compiled by Java Compiler (It does not interact with OS) and converts the java code into bytecode.



What happens at runtime?

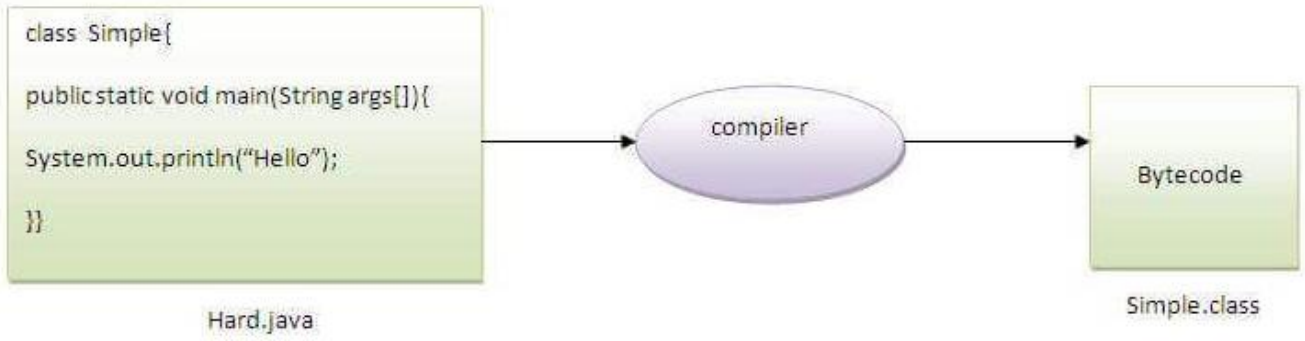
At runtime, following steps are performed:



Classloader: is the subsystem of JVM that is used to load class files.
Bytecode Verifier: checks the code fragments for illegal code that can violate access right to objects.
Interpreter: read bytecode stream then execute the instructions.

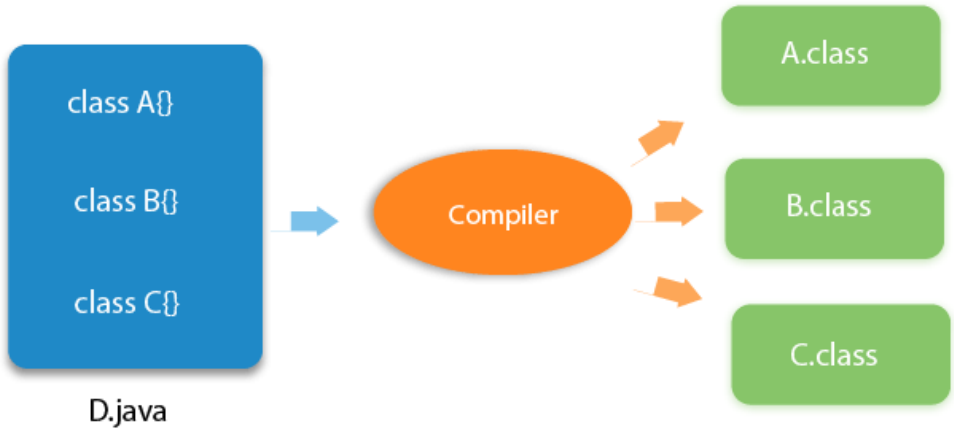
Q) Can you save a java source file by other name than the class name?

Yes, if the class is not public. It is explained in the figure given below:



To compile: javac Hard.java
To execute: java Simple

Q) Can you have multiple classes in a java source file?
Yes, like the figure given below illustrates:



Questions and Answers

1. What is Java?
Answer: Java is a high-level, object-oriented, platform-independent programming language developed by Sun Microsystems (now owned by Oracle). It is known for its portability, security, and robustness.

2. What is the meaning of "Platform" in Java?
Answer: A platform is the environment in which a program runs. Java provides a software-based platform (JRE + API), making it platform-independent.

3. What are the types of Java applications?
Answer:

1. Standalone Applications (e.g., Media Player)
2. Web Applications (e.g., IRCTC)
3. Enterprise Applications (e.g., Banking systems)
4. Mobile Applications (e.g., Android apps)

4. What is bytecode in Java?
Answer: Bytecode is the intermediate code generated by the Java compiler. It is platform-independent and executed by the JVM.

5. What are the key features (buzzwords) of Java?
Answer:

- Simple
- Object-Oriented
- Platform Independent
- Secure
- Robust
- Portable
- High Performance
- Distributed
- Multithreaded
- Dynamic

6. What makes Java platform-independent?
Answer: Java is platform-independent because it compiles code into bytecode, which runs on any device with a JVM.

7. What is the difference between JDK, JRE, and JVM?
Answer:

- JDK: Java Development Kit – includes JRE + development tools
- JRE: Java Runtime Environment – includes JVM + libraries
- JVM: Java Virtual Machine – runs bytecode

8. What is the function of System.out.println() in Java?
Answer:
It prints messages to the console.

- System is a class
- out is a static PrintStream object
- println() is a method to print and go to the next line

9. Can you save a Java file with a name different from the class name?
Answer: Yes, only if the class is not declared public. If the class is public, the file name must match the class name.

10. What happens when a Java program is compiled and executed?
Answer:

- Compile Time: .java file is converted to .class (bytecode)
- Run Time: JVM loads classes, verifies bytecode, and interprets it using the Classloader, Bytecode Verifier, and Interpreter

11. What is the Java Virtual Machine (JVM)?
Answer: The JVM is a part of the Java Runtime Environment (JRE) that runs Java bytecode on any platform. It makes Java platform-independent by converting bytecode into machine code at runtime.

12. Explain the difference between JDK, JRE, and JVM.

Answer:

Term	Description
JDK (Java Development Kit)	Full software package for Java development, includes compiler, debugger, and JRE
JRE (Java Runtime Environment)	Environment to run Java applications, includes JVM and libraries
JVM (Java Virtual Machine)	Part of JRE that runs Java bytecode on the host machine

13. Why is Java platform-independent?

Answer: Java code is compiled into bytecode that can be run on any machine with a JVM, regardless of underlying OS or hardware.

14. What are valid main() method signatures in Java?

Answer:

- public static void main(String[] args)
- public static void main(String... args)
- static public void main(String[] args)

15. What makes Java secure?

Answer:

- No explicit pointers
- ClassLoader separates packages
- Bytecode Verifier checks bytecode
- Security Manager controls access
- Support for cryptography, SSL (Secure Sockets Layer), JAAS (Java Authentication and Authorization Service)

16. What are Java data types?

Answer: Java has two types of data types:

- Primitive: byte, short, int, long, float, double, char, Boolean
    - **byte:** 8-bit signed integer.
    - **short:** 16-bit signed integer.
    - **int:** 32-bit signed integer.
    - **long:** 64-bit signed integer.
    - **float:** 32-bit single-precision floating-point number.
    - **double:** 64-bit double-precision floating-point number.
    - **boolean:** Represents a logical value, either true or false.
    - **char:** 16-bit Unicode character.
- Non-primitive: String, Array, Class, etc.

17. What are variables in Java?

Answer: Variables are containers that store data values. They must be declared with a type:

18. What are the different types of Java variables?

Answer:

- Local Variables – declared inside a method
- Instance Variables – declared inside a class, but outside a method
- Static Variables – declared with static keyword; shared among all instances

19. What is the role of the public static void main(String[] args) method?

Answer:

- public: Accessible from anywhere
- static: No need to create an object to call it
- void: Returns nothing
- main: Name of the entry method
- String[] args: Command-line arguments

20. What is Java IDE? Give examples.

Answer:

An IDE (Integrated Development Environment) is a software suite for writing and testing Java code.

Examples:

- Eclipse
- IntelliJ IDEA
- NetBeans
- BlueJ (for beginners)

21. What is method in Java?

Answer:

A method is a block of code that performs a specific task. It can be reused and called multiple times.

Example:

```
public int add(int a, int b) {
   return a + b;
}
```

22. What is the difference between Java and C++?

Answer:

Feature	Java	C++
Memory Management	Automatic (Garbage Collected)	Manual (Pointers)
Platform Independence	Yes	No
Multiple Inheritance	Not supported directly	Supported
Pointers	Not allowed	Allowed